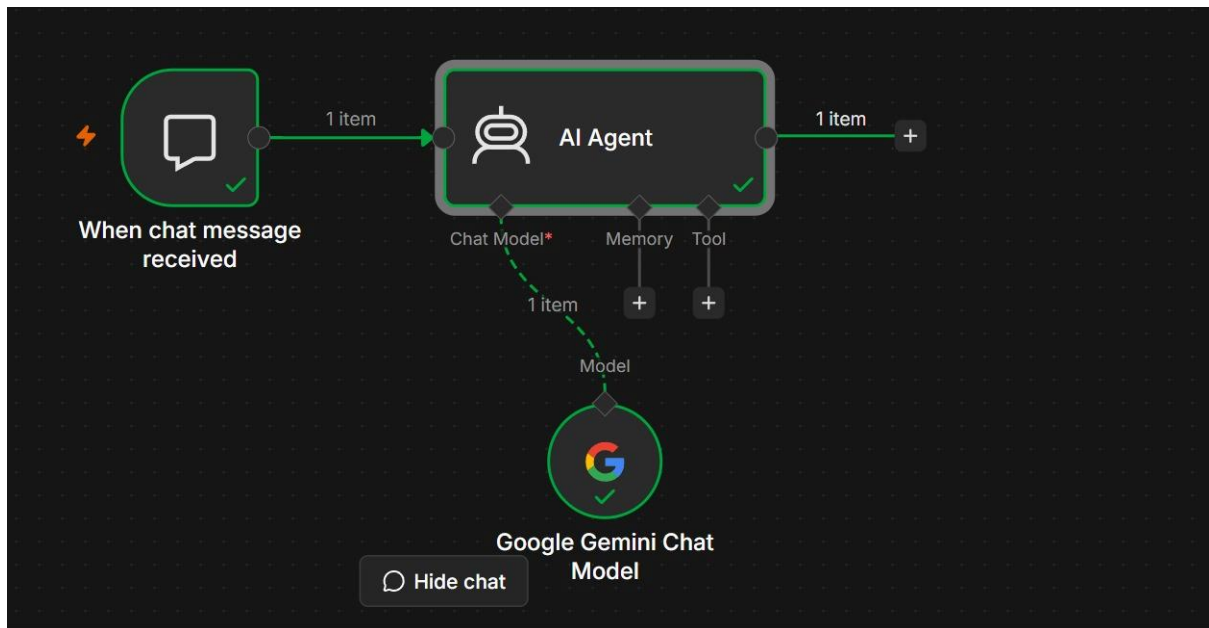


PROJECT-17

AI Tweet Generator



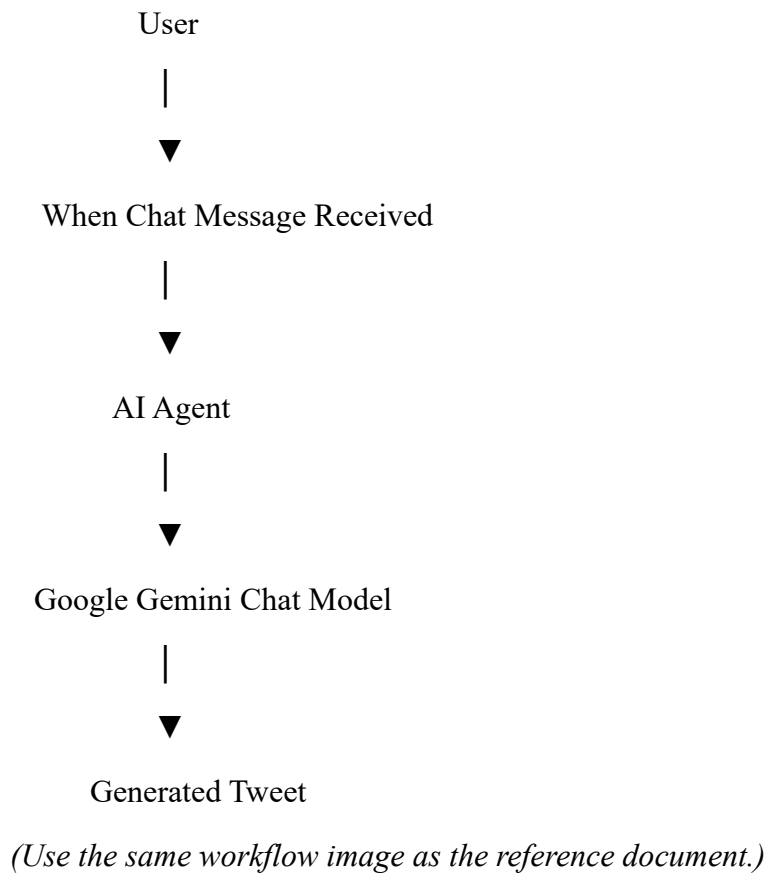
Project Objective

The objective of this project is to build an **AI-powered Tweet Generator** that creates engaging, concise, and impactful tweets based on a user's topic or idea.

The AI Agent can generate:

- Promotional tweets
- Personal branding tweets
- Product launch tweets
- Marketing tweets
- Motivational tweets
- Technology tweets
- Business updates
- Educational tweets
- Event announcements
- Trending social media content

Workflow Architecture



Workflow Components

Step	Node	Purpose
1	When Chat Message Received	Receives the user's request
2	AI Agent	Understands the topic and prepares the prompt
3	Google Gemini Chat Model	Generates an engaging tweet

Prerequisites

Before building this project, ensure you have:

- n8n (Cloud or Self-hosted)
 - Google AI Studio API Key
 - Google Gemini Chat Model configured
 - Internet connection
-

Step 1 – Create Workflow

Open **n8n**

Click

New Workflow

Rename it

AI Tweet Generator

Step 2 – Add Chat Trigger

Add

When Chat Message Received

Purpose

Receives the user's request.

Example Input

Write a tweet announcing our new AI automation platform.

Output

```
{  
  "chatInput": "Write a tweet announcing our new AI automation platform."  
}
```

Step 3 – Add AI Agent

Connect

When Chat Message Received

|



AI Agent

Configure

Source for Prompt

Connected Chat Trigger Node

Prompt

```
{{ $json.chatInput }}
```

Step 4 – Configure the System Prompt

Paste the following into the **System Message**.

You are an AI Tweet Generator.

Your task is to create engaging, concise, and impactful tweets.

Instructions:

- Understand the user's topic.
 - Write tweets that are clear, engaging, and suitable for X (Twitter).
 - Keep the tweet within 280 characters unless the user requests a thread.
 - Use emojis only when appropriate.
 - Include relevant hashtags.
 - Create a strong hook that captures attention.
 - Maintain a professional yet conversational tone.
 - Return only the tweet.
-

Step 5 – Add Google Gemini Chat Model

Click

+ **Chat Model**

Choose

Google Gemini Chat Model

Configuration

Setting	Value
Provider	Google Gemini
Model	Gemini 2.5 Flash (or another available Gemini model)

Setting	Value
---------	-------

Authentication	Google AI Studio API Key
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Connect it to the AI Agent.

Final Workflow

When Chat Message Received

|



AI Agent

|



Google Gemini Chat Model

Workflow Execution

Step 1

User enters

Write a tweet about learning AI automation with n8n.

↓

Step 2

Chat Trigger receives the request.

↓

Step 3

AI Agent understands the request.

↓

Step 4

The prompt is sent to the Google Gemini Chat Model.

↓

Step 5

Gemini generates a concise, engaging tweet.

↓

Step 6

The tweet is returned to the user.

Sample Inputs and Outputs

Example 1

Input

Write a tweet about learning AI.

Output

 Every day is a new opportunity to learn AI and build something amazing. Start small, stay consistent, and keep experimenting!

#AI #MachineLearning #Tech

Example 2

Input

Create a product launch tweet.

Output

 Introducing our newest AI automation platform! Build smarter workflows, save time, and boost productivity with the power of AI.

#AI #Automation #Innovation

Example 3

Input

Write a motivational tweet for developers.

Output

 Every bug you fix and every project you complete makes you a stronger developer. Keep building, keep learning!

#Coding #Developers #Programming

Example 4

Input

Create a marketing tweet for an online course.

Output

🧑🏻💻 Ready to level up your AI skills? Join our hands-on course and learn to build real-world AI workflows with n8n and Gemini!

#AI #Learning #Automation

Workflow Diagram

User

|



When Chat Message Received

|



AI Agent

|



Google Gemini Chat Model

|



Generated Tweet

Error Handling

Error: No Prompt Specified

Cause

The AI Agent did not receive a user request.

Solution

Verify that the Prompt field contains:

```
{{ $json.chatInput }}
```

Error: Authentication Failed

Cause

Invalid Google Gemini API Key.

Solution

Reconnect or update your Google AI Studio credentials.

Error: Too Many Requests

Cause

Gemini API rate limit exceeded.

Solution

- Wait a few minutes.
 - Check your API quota.
 - Retry the request.
-

Learning Outcomes

After completing this project, you will be able to:

- Build an AI-powered Tweet Generator in n8n.
 - Configure an AI Agent with Google Gemini.
 - Design prompts for short-form social media content.
 - Generate engaging tweets automatically.
 - Build conversational AI workflows.
-

Real-World Applications

- Social media marketing

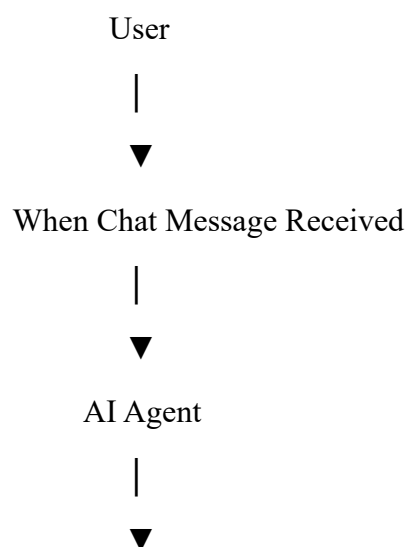
- Brand promotion
 - Product launches
 - Personal branding
 - Company announcements
 - Event promotion
 - Technology updates
 - Educational content
 - Startup marketing
 - Community engagement
-

Practice Challenge

Enhance this project by adding:

1. A tweet tone selector (Professional, Casual, Funny, Inspirational).
 2. A tweet length selector (Short, Standard, Thread).
 3. Automatic hashtag generation based on the topic.
 4. A Memory node to remember the user's preferred writing style.
 5. A Google Sheets node to save generated tweets.
 6. An X (Twitter) API node (or HTTP Request node) to publish tweets automatically.
-

Final Workflow



Google Gemini Chat Model

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Professional Tweet

Expected Result

The **AI Tweet Generator** accepts a topic or idea from the user, uses the **Google Gemini Chat Model** to generate an engaging, concise, and high-quality tweet, and returns content optimized for **X (Twitter)**. This project teaches prompt engineering, AI-powered content generation, and conversational workflow design in n8n, providing a strong foundation for building advanced social media automation agents.